

R for Data Analysis & Visualization



A 10-Week Live Cohort Bootcamp
From R Fundamentals to Reproducible
Analytics & Interactive Dashboards

Delivered by **SI Analytics**

Build real-world analytics, visualization and dashboarding skills using
R, tidyverse, ggplot2, Plotly and Quarto

TABLE OF CONTENTS

03	Bootcamp Overview
04	Bootcamp Features
05	Learning Structure
06	Course Modules Overview
12	Grading & Assessment Rubric
13	Why SI Analytics?
14	Instructors & Mentors

BOOTCAMP OVERVIEW

R for Data Analysis and Visualization is a structured, mentor-guided **10-week live cohort program** designed to take learners from foundational R programming concepts to **professional-grade data analysis, visualization, and reproducible reporting**.

This bootcamp emphasizes:

- Practical, hands-on learning
- Real datasets and applied analytics
- Strong data storytelling and visualization principles
- Reproducible research workflows using modern R tools

Participants graduate with **job-ready skills** and a **public portfolio of analytics projects** built entirely in R, demonstrating real-world competence and analytical depth

BOOTCAMP FEATURES



Weekly live
instructor-led
sessions



Structured pre-work
and guided code-
along exercises



Hands-on workshop
assignments with
details feedback



Mentor-guided
practical and problem
solving sessions



Private learning
community and
peer collaboration



Real-world datasets
and applied case
studies



Capstone analytics
and interactive
dashboard
project using R



Shareabel SI
Analytics Certificate
of Completion



Lifetime access to
curated learning
resources

► Duration
10 Weeks

► Format
Live + Guided Self-Paced Learning

LEARNING STRUCTURE

How the Bootcamp Works

Each week of the R for Data Analysis and Visualization bootcamp follows a consistent, learner-friendly structure designed to balance flexibility with accountability and steady progress.

01

Pre-work (2–3 hours)

Pre-recorded videos, guided readings, and structured code-along exercises to prepare learners for the live session.

02

Live Workshop (2 hours)

Instructor-led deep dives with hands-on practice, real-time problem solving, and guided demonstrations

03

Post-Workshop Assignment

Applied task submitted for review and feedback to reinforce learning and build practical competence

04

Optional Support Session

Alternate live session for learners who cannot attend the primary workshop, ensuring inclusivity and continuity.

This structure balances **flexibility with accountability**, ensuring steady progress and meaningful skill development throughout the program

COURSE MODULES OVERVIEW

10-Week Structured Learning Journey

The R for Data Analysis & Visualization bootcamp is organized into a progressive sequence of modules designed to build strong analytical foundations and advance learners toward professional-grade, reproducible analytics and interactive dashboards using R.

MODULE 1

**R Foundations for
Data Analysis**

MODULE 2

**Data Visualization
with ggplot2**

MODULE 3

**Data Wrangling &
Analysis with tidyverse**

MODULE 4

**Reproducible Reporting
with Quarto**

MODULE 5

**Dashboards
with Quarto**

MODULE 6

**Capstone Project &
Portfolio Deployment**

MODULE 1

R Foundations

This module establishes a strong foundation in R programming and analytical thinking. Learners are introduced to core R concepts, data structures, and development environments that underpin professional data analysis workflows.

Key Topics

- R syntax and core programming concepts
- Data types, vectors, lists, and data frames
- Control structures and functions
- Working in RStudio and VS Code
- Introduction to Quarto for reproducible analysis
- Exploratory data analysis concepts
- Responsible use of AI tools for coding support

Outcome: Learners gain confidence writing clean, readable **R code** for data analysis and building reproducible analytical workflows.

MODULE 2

Data Visualization with ggplot2

This module focuses on communicating insights through clear, elegant, and effective data visualizations. Learners master the Grammar of Graphics framework and apply best practices for visual storytelling using R.

KEY TOPICS

- Data visualization principles and storytelling
- Grammar of Graphics using ggplot2
- Custom themes, color palettes, and layout design
- Faceting, annotations, and informative labels
- Building publication-ready and presentation-quality charts

Outcome: Learners create polished, professional visualizations suitable for reports, academic publications, and executive presentations.

MODULE 3

Data Wrangling with tidyverse

This module equips learners with the skills to clean, transform, and analyze real world datasets using modern, readable, and reproducible R workflows built on the tidyverse ecosystem.

KEY TOPICS

- Data cleaning and preprocessing
- Filtering, grouping, and summarization
- Feature engineering and transformations
- Handling missing data and data types
- Joining, reshaping, and tidying datasets

Outcome: Learners confidently manipulate complex datasets using **tidyverse workflows** to support analysis, visualization, and reproducible reporting.

MODULE 4

Reproducible Reporting with Quarto

This module introduces **reproducible research and reporting workflows** using **Quarto**. Learners design structured analytical documents that seamlessly integrate R code, data, tables, and visualizations into shareable outputs.

KEY TOPICS

- Quarto documents, projects, and directory structure
- Embedding R code, tables, and plots
- Parameterized reports for reusable analytics
- Styling, formatting, and layout customization
- Publishing reports and dashboards to GitHub Pages

Outcome: Learners produce fully **reproducible, transparent, and shareable analytical reports** suitable for professional, academic, and organizational use.

MODULE 5 & CAPSTONE

Dashboards with Quarto

This module focuses on building professional, interactive dashboards using Quarto in R. Learners integrate data analysis, visualization, and narrative storytelling to design dashboards suitable for web publishing and decision-making contexts.

KEY TOPICS

- Quarto fundamentals for dashboards
- Building interactive dashboards with Quarto
- Theming and branding analytical applications
- Deploying Quarto dashboards to GitHub Pages

CAPSTONE PROJECT

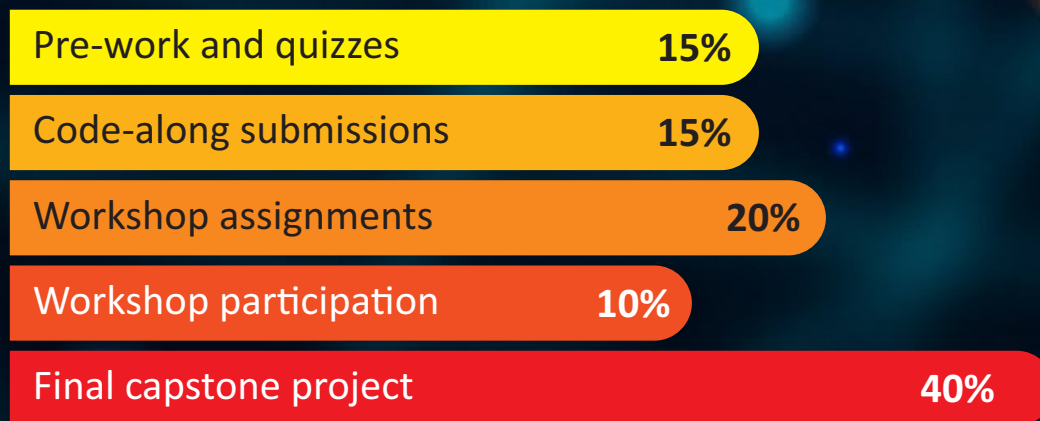
Participants design and deploy a **fully interactive R-based analytics dashboard**, demonstrating end-to-end proficiency from data wrangling and analysis to visualization, storytelling, and deployment.

Outcome: A public, shareable analytics portfolio piece built entirely in **R**, showcasing professional-grade dashboarding and analytical skills.

GRADING & ASSESSMENT

Grading Rubric

Final performance in the R for Data Analysis and Visualization bootcamp is assessed using a balanced combination of participation, practical assignments, and a comprehensive capstone project.



Passing Requirement: Learners must achieve a minimum overall score of **80%** to successfully complete the bootcamp and receive the SI Analytics Certificate of Completion

WHY SI ANALYTICS?

SI Analytics is a global analytics education and innovation hub committed to **quality, rigor, and practical impact**. We design learning experiences that bridge theory and practice, empowering professionals to apply analytics confidently in real-world contexts.

Expert-led instruction
grounded in real-world analytics

Strong emphasis on
visualization and storytelling

Mentor-guided learning and a
supportive global community

Proven track record across
public health, business, and
development sectors

Learner-centered, outcomes
driven program design

We do not just teach tools.

We build analytical thinkers and problem solvers.

INSTRUCTORS AND MENTORS

Dr. Olugbenga Asaolu

Strategic Evidence &
Data Science Leader

[Linkedin Profile](#)

Dr. Elysee Yuyishime

Data Scientist & Global
Health Researcher

[Linkedin Profile](#)

Michael Sileshi

Data Analyst & Visualization
Specialist

[Linkedin Profile](#)

Camila Boynton

Data Visualization &
Analytics Educator

[Linkedin Profile](#)

Tayo Asaolu

Analytics & Learning
Operations Lead

[Linkedin Profile](#)



www.sianalytics.org | contact@sianalytics.org

LinkedIn: Strategic Insights and Analytics (SI Analytics)

Facebook: Strategic Insights and Analytics (SI Analytics)

YouTube: SIAnalyticsLab

X (Twitter): SIAnalyticsLab